Problems Set 3: Departure from Assumptions of the Normal Linear Model

Detailed, Self-Explanatory Step-by-Step Solutions

BSDS V: Regression Techniques

Prepared from the supplied Problems Set 3, the uploaded Regression Techniques lecture-note source, and the structure of the supplied PS1/PS2 solution sets.

How to use this solution set

Each question follows the same sequence used in the supplied solution sets:

1. the complete assignment question, transcribed before solving;
2. the definitions, identities, and theory needed for that question, drawn from the supplied lecture notes;
3. a step-by-step derivation, with the algebraic transition explained;
4. a boxed conclusion, including any qualification needed for the result to be mathematically correct.

Notation and standing assumptions

Throughout PS3, the design matrix contains an intercept and p regressors:

$$X = [\,\mathbf{1}\;\; x^{(1)}\;\; \cdots\;\; x^{(p)}\,] \in \mathbb{R}^{n\times(p+1)}, \qquad Y = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

Unless a question says otherwise, assume that X has full column rank $p+1$. Define

$$P_X = X(X^TX)^{-1}X^T, \qquad M_X = I_n - P_X, \qquad \nu = n - p - 1.$$

Thus P_X projects onto the column space of X, while M_X projects onto its orthogonal complement. In particular,

$$P_X^T = P_X, \quad P_X^2 = P_X, \quad M_X^T = M_X, \quad M_X^2 = M_X, \quad \operatorname{rank}(P_X) = p+1, \quad \operatorname{rank}(M_X) = \nu.$$

When normal-theory intervals or residual distributions are used, the normality assumption stated in the relevant question is also imposed. Vectors are columns unless explicitly described as rows.

Contents

1 Question 1: Adding unnecessary predictors and interval widths **3**

2 Question 2: Heteroskedasticity and the OLS variance estimator **6**

3 Question 3: Weighted least squares when the heteroskedasticity is known **8**

4 Question 4: Expected Mahalanobis distances and hat-matrix entries **10**

5 Question 5: Extra sum of squares and the covariance of the added coefficients 13

6 Question 6: Deleted residuals and the deleted-sample hat matrix 15

7 Question 7: Increase in RSS after adding one observation 18

8 Question 8: An unbiased estimator under equicorrelated errors 19

9 Question 9: Covariance matrix and distribution of AR(1) errors 21

10 Question 10: Effect of scaling on VIF and condition number 22

11 Question 11: Projection representation and Beta law of a standardized residual 24

12 Question 12: Mean and variance of standardized residuals 27

13 Question 13: Sensitivity of a contrast to a response perturbation 29

14 Question 14: Determinant ratio after deleting multiple observations 31

End of Problems Set 3 solutions 32

1 Question 1: Adding unnecessary predictors and interval widths

Question (as assigned)

1. Let the underlying actual model be (1). However, you have collected data on q additional variables $\{z_1, \ldots, z_q\}$ and work on the model:

$$Y = X\beta + Z\gamma + \epsilon^\star, \qquad \epsilon^\star \sim N(0, \sigma^2 I), \qquad Z = [\, z^{(1)} \ \cdots \ z^{(q)} \,]. \tag{2}$$

Consider a new observation $(x_0^T : z_0^T)$ with corresponding response Y_0. Compare the confidence interval of $x_0^T\beta$ obtained from your model (4) with the one obtained from the actual model (1). Also, compare the prediction interval for Y_0 obtained from model (4) with the one obtained from the actual model (1).

Interpretation note about the printed cross-reference

The question first says that model (1), $Y = X\beta + \epsilon$, is the actual model, and then says that the fitted model includes the additional columns Z as in (2). The phrase "model (4)" is a cross-reference inconsistency: model (4) is not introduced until Question 5, where it is the reduced X-only model. The derivation below follows the explicit setup in Question 1 and the lecture-note section on *variance inflation from over-specification*: the true model is $Y = X\beta + \epsilon$, while the fitted model is the enlarged model $Y = X\beta + Z\gamma + \epsilon^\star$, with the true extra coefficient equal to zero. No change to the quoted question is made silently.

Required theory from the supplied lecture notes

OLS covariance under a correctly specified spherical-error model. If $Y = A\theta + \epsilon$, $\mathbb{E}(\epsilon) = 0$, and $\mathrm{Var}(\epsilon) = \sigma^2 I$, then

$$\widehat{\theta} = (A^T A)^{-1} A^T Y, \qquad \mathbb{E}(\widehat{\theta}) = \theta, \qquad \mathrm{Var}(\widehat{\theta}) = \sigma^2 (A^T A)^{-1}.$$

For any fixed vector c, the variance of the estimated linear combination is $c^T \mathrm{Var}(\widehat{\theta}) c$.

Nested column spaces and the Schur complement. Put $W = [X \ Z]$ and

$$H = Z^T (I - P_X) Z.$$

If W has full column rank, then H is positive definite and the block inverse formula gives

$$(W^T W)^{-1}_{11} = (X^T X)^{-1} + (X^T X)^{-1} X^T Z H^{-1} Z^T X (X^T X)^{-1}.$$

The added matrix is positive semidefinite because it has the form $AH^{-1}A^T$.

Confidence and prediction intervals. Under normal errors, a confidence interval for a regression mean $c^T\theta$ has the form

$$\widehat{c^T\theta} \ \pm \ t_{\nu, 1-\alpha/2}\, s \sqrt{c^T (A^T A)^{-1} c}, \qquad s^2 = \frac{\mathrm{RSS}}{\nu}.$$

For a new independent response at design row a_0^T, the prediction standard error contains the extra future-noise variance:

$$s \sqrt{1 + a_0^T (A^T A)^{-1} a_0}.$$

The 1 is absent for a confidence interval for the mean and present for a prediction interval for an individual future response.

What the comparison will show. Adding columns that are unnecessary in the true model does not bias OLS, but it increases or leaves unchanged the variance of the estimated coefficients and predictions. We will obtain the exact nonnegative extra terms, rather than relying only on a verbal statement.

Step 1: Write the true and fitted models separately

Under the interpretation specified above, the data-generating model is

$$Y = X\beta + \epsilon, \qquad \epsilon \sim N_n(0, \sigma^2 I_n).$$

The fitted model uses the enlarged design matrix

$$W = [X\ Z], \qquad \delta = \begin{pmatrix} \beta \\ 0_q \end{pmatrix}, \qquad Y = W\delta + \epsilon.$$

The extra coefficients are zero in the true model. They are included in the fit, so their estimates fluctuate around zero.

Why this step leads to the next step

This separation matters. The comparison is not between two different true data-generating mechanisms; it is between fitting the correct smaller model and fitting a larger model containing irrelevant regressors.

Step 2: Set up the block cross-product matrix

Let

$$A = X^T X, \qquad B = X^T Z, \qquad C = Z^T Z, \qquad H = C - B^T A^{-1} B = Z^T (I - P_X) Z.$$

Then

$$W^T W = \begin{pmatrix} A & B \\ B^T & C \end{pmatrix}.$$

The block inverse formula yields

$$(W^T W)^{-1} = \begin{pmatrix} A^{-1} + A^{-1} B H^{-1} B^T A^{-1} & -A^{-1} B H^{-1} \\ -H^{-1} B^T A^{-1} & H^{-1} \end{pmatrix}.$$

The top-left block is the covariance multiplier for the estimator of β when the extra columns are fitted.

Step 3: Compare the covariance matrices for $\widehat{\beta}$

Under the actual X-only model,

$$\widehat{\beta}_X = (X^T X)^{-1} X^T Y, \qquad \mathrm{Var}(\widehat{\beta}_X) = \sigma^2 A^{-1}.$$

Under the enlarged model, let $\widehat{\beta}_W$ denote the first $p+1$ coordinates of $\widehat{\delta} = (W^T W)^{-1} W^T Y$. Its covariance matrix is

$$\begin{aligned} \mathrm{Var}(\widehat{\beta}_W) &= \sigma^2 \left[A^{-1} + A^{-1} B H^{-1} B^T A^{-1} \right] \\ &= \sigma^2 A^{-1} + \sigma^2 A^{-1} X^T Z H^{-1} Z^T X A^{-1}. \end{aligned}$$

For every vector $v \in \mathbb{R}^{p+1}$,

$$v^T \left(\mathrm{Var}(\widehat{\beta}_W) - \mathrm{Var}(\widehat{\beta}_X) \right) v = \sigma^2 (B^T A^{-1} v)^T H^{-1} (B^T A^{-1} v) \geq 0.$$

Therefore

$$\mathrm{Var}(\widehat{\beta}_W) - \mathrm{Var}(\widehat{\beta}_X) \succeq 0.$$

Why this step leads to the next step

The matrix inequality is stronger than saying that a few diagonal variances increase: it says that the variance of *every* fixed linear combination of the coefficient estimates cannot decrease when the unnecessary columns are added.

Step 4: Compare the confidence intervals for $x_0^T\beta$

The actual-model estimator of the regression mean at x_0 is $x_0^T\widehat{\beta}_X$. Its variance is

$$V_X(x_0) = \sigma^2 x_0^T A^{-1} x_0.$$

The enlarged-model estimator of the same target is $x_0^T\widehat{\beta}_W$. Define

$$a_0 = Z^T X A^{-1} x_0 \in \mathbb{R}^q.$$

Using the covariance expression from Step 3,

$$\begin{aligned} V_W(x_0) &= \sigma^2 x_0^T \left(A^{-1} + A^{-1}BH^{-1}B^T A^{-1}\right) x_0 \\ &= \sigma^2 x_0^T A^{-1} x_0 + \sigma^2 a_0^T H^{-1} a_0 \\ &\geq V_X(x_0). \end{aligned}$$

Thus, if σ is treated as known, the enlarged-model confidence interval has at least as large a standard error and hence is no shorter. Equality holds exactly when $a_0 = 0$.

For the usual unknown-variance intervals, the two conventional forms are

$$\begin{aligned} CI_X: &\quad x_0^T\widehat{\beta}_X \;\pm\; t_{n-p-1,1-\alpha/2}\, s_X \sqrt{x_0^T A^{-1} x_0}, \\ CI_W: &\quad x_0^T\widehat{\beta}_W \;\pm\; t_{n-p-q-1,1-\alpha/2}\, s_W \sqrt{x_0^T A^{-1} x_0 + a_0^T H^{-1} a_0}, \end{aligned}$$

where

$$s_X^2 = \frac{Y^T(I - P_X)Y}{n-p-1}, \qquad s_W^2 = \frac{Y^T(I - P_W)Y}{n-p-q-1}.$$

Both variance estimators are unbiased under the stated true model because the additional true coefficients are zero. The enlarged model has fewer residual degrees of freedom and a larger theoretical variance multiplier. However, s_X and s_W are random quantities, so one cannot claim that every realized plug-in interval must be longer for every possible dataset. The guaranteed comparison is the covariance/standard-error component displayed above (and the interval using known σ).

Step 5: Compare the prediction intervals for a future response

For the actual model,

$$Y_0 = x_0^T\beta + \epsilon_0, \qquad \epsilon_0 \sim N(0, \sigma^2),$$

with ϵ_0 independent of the training errors. The actual-model predictor is $\widehat{Y}_{0,X} = x_0^T\widehat{\beta}_X$, so

$$\mathrm{Var}(Y_0 - \widehat{Y}_{0,X}) = \sigma^2\left(1 + x_0^T A^{-1} x_0\right).$$

The enlarged-model predictor at (x_0, z_0) is

$$\widehat{Y}_{0,W} = \begin{pmatrix} x_0^T & z_0^T \end{pmatrix}\widehat{\delta}.$$

Using the full block inverse and defining $a_0 = Z^T X A^{-1} x_0$ as above gives

$$\begin{aligned}&\begin{pmatrix} x_0^T & z_0^T \end{pmatrix} (W^T W)^{-1} \begin{pmatrix} x_0 \\ z_0 \end{pmatrix} \\ &\qquad = x_0^T A^{-1} x_0 + (z_0 - a_0)^T H^{-1} (z_0 - a_0).\end{aligned}$$

Consequently,

$$\mathrm{Var}(Y_0 - \widehat{Y}_{0,W}) = \sigma^2 \left[1 + x_0^T A^{-1} x_0 + (z_0 - a_0)^T H^{-1} (z_0 - a_0)\right].$$

The final term is nonnegative. It is zero only if $z_0 = a_0$. Thus the enlarged-model prediction variance cannot be smaller than the actual-model prediction variance.

The corresponding usual prediction intervals are

$$\begin{aligned} PI_X: &\quad \widehat{Y}_{0,X} \pm t_{n-p-1,1-\alpha/2} s_X \sqrt{1 + x_0^T A^{-1} x_0}, \\ PI_W: &\quad \widehat{Y}_{0,W} \pm t_{n-p-q-1,1-\alpha/2} s_W \sqrt{1 + x_0^T A^{-1} x_0 + (z_0 - a_0)^T H^{-1} (z_0 - a_0)}. \end{aligned}$$

As for the confidence intervals, the theoretical prediction variance has a nonnegative extra term; the realized plug-in interval lengths also depend on the random residual variance estimates.

Final answer

Adding irrelevant predictors preserves unbiasedness when their true coefficients are zero, but inflates (or leaves unchanged) the covariance matrix of $\widehat{\beta}$. In particular,

$$\boxed{\mathrm{Var}(\widehat{\beta}_W) - \mathrm{Var}(\widehat{\beta}_X) = \sigma^2 A^{-1} X^T Z H^{-1} Z^T X A^{-1} \succeq 0.}$$

For the mean at x_0, the extra variance is $\sigma^2 a_0^T H^{-1} a_0 \geq 0$. For prediction at (x_0, z_0), the extra variance beyond the actual-model prediction variance is $\sigma^2 (z_0 - a_0)^T H^{-1} (z_0 - a_0) \geq 0$. Therefore the over-specified model has no theoretical interval-width advantage; with estimated σ^2, individual realized widths need not be ordered deterministically.

2 Question 2: Heteroskedasticity and the OLS variance estimator

Question (as assigned)

2. Consider the linear model in (1) as your working model. Suppose, however, that the underlying true model is as in (1) except for the fact that

$$\mathrm{var}(\epsilon) = \sigma^2 \,\mathrm{Diag}(t, t^2, \ldots, t^n)$$

for some number $t > 1$. How would this misspecification affect the variance of OLS $\widehat{\beta}_{OLS}$ and expectation of $\widehat{\sigma}^2_{UE} = \mathrm{RSS}/(n - p - 1)$?

Required theory from the supplied lecture notes

The supplied notes give, for $Y = X\beta + \epsilon$ with $\mathbb{E}(\epsilon) = 0$ and $\mathrm{Var}(\epsilon) = \sigma^2 V$,

$$\widehat{\beta}_{OLS} = (X^T X)^{-1} X^T Y, \qquad \mathrm{Var}(\widehat{\beta}_{OLS}) = \sigma^2 (X^T X)^{-1} X^T V X (X^T X)^{-1}.$$

They also use the projection expression

$$\mathrm{RSS} = Y^T (I - P_X) Y.$$

For a zero-mean random vector U and a fixed matrix A,

$$\mathbb{E}(U^T A U) = \mathrm{tr}\{A \,\mathrm{Var}(U)\}.$$

The residual-maker $M_X = I - P_X$ is symmetric and idempotent, with $\operatorname{tr}(M_X) = n - (p+1) = n - p - 1$.

What we need to calculate. We must compute the actual OLS covariance under the diagonal covariance matrix, and then compute the expectation of the usual homoskedastic residual variance estimator under that same true covariance.

Step 1: Write the heteroskedastic covariance matrix

Let

$$D = \operatorname{Diag}(t, t^2, \ldots, t^n).$$

The true model is still $Y = X\beta + \epsilon$ with mean zero errors, but now

$$\operatorname{Var}(\epsilon) = \sigma^2 D.$$

Since $t > 1$, every diagonal entry of D is greater than 1; hence $D - I_n$ is positive definite.

Step 2: Find the actual covariance of the OLS estimator

Set $A = X^T X$. The OLS estimator remains

$$\widehat{\beta}_{OLS} = A^{-1} X^T Y.$$

It is still unbiased because

$$\mathbb{E}(\widehat{\beta}_{OLS}) = A^{-1} X^T \mathbb{E}(Y) = A^{-1} X^T X \beta = \beta.$$

Its covariance, however, must be evaluated using the true covariance $\sigma^2 D$:

$$\boxed{\operatorname{Var}(\widehat{\beta}_{OLS}) = \sigma^2 (X^T X)^{-1} X^T D X (X^T X)^{-1}.}$$

Step 3: Compare it with the covariance under homoskedasticity

If the errors had covariance $\sigma^2 I_n$, the familiar covariance would be $\sigma^2 A^{-1}$. The difference under the present true model is

$$\begin{aligned} &\operatorname{Var}(\widehat{\beta}_{OLS}) - \sigma^2 A^{-1} \\ &\quad = \sigma^2 A^{-1} X^T (D - I_n) X A^{-1}. \end{aligned}$$

For any $c \in \mathbb{R}^{p+1}$,

$$c^T A^{-1} X^T (D - I_n) X A^{-1} c = (X A^{-1} c)^T (D - I_n)(X A^{-1} c) \geq 0.$$

Therefore the covariance difference is positive semidefinite. Since $D - I_n$ is positive definite and X has full column rank, it is in fact positive definite for nonzero c.

Why this step leads to the next step

The OLS formula itself has not changed, and it remains unbiased. What has changed is the variability of the errors entering that formula. The diagonal weights $t, t^2, \ldots, t^n$ make the actual coefficient covariance larger than the covariance formula obtained by incorrectly assuming equal error variances.

Step 4: Find the expectation of RSS

Let $M_X = I_n - P_X$. Since $M_X X = 0$,

$$M_X Y = M_X(X\beta + \epsilon) = M_X \epsilon,$$

and thus

$$\mathrm{RSS} = Y^T M_X Y = \epsilon^T M_X \epsilon.$$

Using the quadratic-form expectation identity,

$$\begin{aligned}\mathbb{E}(\mathrm{RSS}) &= \mathrm{tr}\{M_X \,\mathrm{Var}(\epsilon)\} \\ &= \sigma^2 \,\mathrm{tr}(M_X D).\end{aligned}$$

Because D is diagonal,

$$\mathrm{tr}(M_X D) = \sum_{i=1}^{n} t^i (M_X)_{ii} = \sum_{i=1}^{n} t^i (1 - p_{ii}).$$

It follows that

$$\mathbb{E}(\widehat{\sigma}^2_{UE}) = \frac{\sigma^2}{n-p-1} \sum_{i=1}^{n} t^i (1 - p_{ii}).$$

Step 5: Determine the direction of the bias

Use $\mathrm{tr}(M_X) = \sum_i (1 - p_{ii}) = n - p - 1$. Then

$$\begin{aligned}\mathbb{E}(\widehat{\sigma}^2_{UE}) - \sigma^2 &= \frac{\sigma^2}{n-p-1} \sum_{i=1}^{n} (t^i - 1)(1 - p_{ii}) \\ &\geq 0,\end{aligned}$$

since $t^i - 1 > 0$ and $1 - p_{ii} \geq 0$. The inequality is strict whenever at least one observation contributes a nonzero residual-space diagonal element, which holds when the residual degrees of freedom are positive.

Final answer

The OLS estimator remains unbiased, but its true covariance is

$$\boxed{\mathrm{Var}(\widehat{\beta}_{OLS}) = \sigma^2 (X^T X)^{-1} X^T D X (X^T X)^{-1}, \quad D = \mathrm{Diag}(t, t^2, \ldots, t^n).}$$

This exceeds the homoskedastic covariance $\sigma^2 (X^T X)^{-1}$ in the positive-semidefinite order. Also,

$$\boxed{\mathbb{E}(\widehat{\sigma}^2_{UE}) = \frac{\sigma^2}{n-p-1} \sum_{i=1}^{n} t^i (1 - p_{ii}) > \sigma^2.}$$

Thus the usual OLS residual variance estimator overestimates the common scale parameter σ^2 under the stated increasing heteroskedasticity.

3 Question 3: Weighted least squares when the heteroskedasticity is known

Question (as assigned)

> 3. Consider the above setup in Q.2. However, suppose that you know t. How would you modify the regression estimate of β so that its variance is smaller than the variance of $\widehat{\beta}_{OLS}$ obtained in Q.2 (in the sense that the difference of the two covariance matrices is negative semi-definite)?

Required theory from the supplied lecture notes

The GLS/WLS part of the supplied notes considers

$$Y = X\beta + \epsilon, \qquad \epsilon \sim N(0, \sigma^2\Sigma),$$

with known positive-definite Σ. Multiplying by $\Sigma^{-1/2}$ gives a transformed model with spherical errors. Its least-squares estimator is

$$\widehat{\beta}_{GLS} = (X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1}Y,$$

with covariance

$$\mathrm{Var}(\widehat{\beta}_{GLS}) = \sigma^2(X^T\Sigma^{-1}X)^{-1}.$$

For diagonal Σ, this is weighted least squares: observations with smaller error variance receive larger weight. The Gauss–Markov comparison says that among linear unbiased estimators, GLS has minimum covariance in the positive-semidefinite ordering.

What changes from Q2. In Q2, the error covariance was known up to the common scale σ^2: it was $\sigma^2 D$. Since t is now known, we can use D^{-1} as the observation-weight matrix.

Step 1: Define the known weight matrix

Let

$$D = \mathrm{Diag}(t, t^2, \ldots, t^n), \qquad D^{-1} = \mathrm{Diag}(t^{-1}, t^{-2}, \ldots, t^{-n}).$$

The covariance is $\sigma^2 D$, so whitening uses $D^{-1/2}$:

$$D^{-1/2}Y = D^{-1/2}X\beta + D^{-1/2}\epsilon, \qquad \mathrm{Var}(D^{-1/2}\epsilon) = \sigma^2 I_n.$$

Step 2: Minimize the transformed sum of squares

OLS on the transformed model minimizes

$$\|D^{-1/2}(Y - X\beta)\|^2 = (Y - X\beta)^T D^{-1}(Y - X\beta).$$

Differentiating this quadratic with respect to β gives the weighted normal equations

$$X^T D^{-1} X\, \widehat{\beta}_{WLS} = X^T D^{-1} Y.$$

Therefore

$$\boxed{\widehat{\beta}_{WLS} = (X^T D^{-1} X)^{-1} X^T D^{-1} Y.}$$

Step 3: Compute its variance

The estimator is linear in Y and is unbiased because

$$\mathbb{E}(\widehat{\beta}_{WLS}) = (X^T D^{-1} X)^{-1} X^T D^{-1} X\beta = \beta.$$

Its covariance is

$$\begin{aligned}\mathrm{Var}(\widehat{\beta}_{WLS}) &= (X^T D^{-1} X)^{-1} X^T D^{-1} (\sigma^2 D) D^{-1} X (X^T D^{-1} X)^{-1} \\ &= \sigma^2 (X^T D^{-1} X)^{-1}.\end{aligned}$$

Step 4: Prove that the covariance is no larger than the OLS covariance

It is not enough merely to state that WLS is appropriate; we must prove the requested matrix inequality. Define the two linear-estimator matrices

$$C = (X^T X)^{-1} X^T, \qquad B = (X^T D^{-1} X)^{-1} X^T D^{-1}.$$

Then $\widehat{\beta}_{OLS} = CY$ and $\widehat{\beta}_{WLS} = BY$. Both are unbiased, so

$$CX = I_{p+1}, \qquad BX = I_{p+1}, \qquad (C - B)X = 0.$$

Now use $\mathrm{Var}(Y) = \sigma^2 D$. Since

$$BDB^T = (X^T D^{-1} X)^{-1},$$

and

$$BD(C - B)^T = (X^T D^{-1} X)^{-1} X^T (C - B)^T = (X^T D^{-1} X)^{-1}((C - B)X)^T = 0,$$

the cross-covariance terms vanish. Expanding the covariance of $(C - B)Y + BY$ gives

$$\mathrm{Var}(\widehat{\beta}_{OLS}) = \mathrm{Var}(\widehat{\beta}_{WLS}) + \sigma^2 (C - B) D (C - B)^T.$$

The last matrix is positive semidefinite: for every v,

$$v^T (C - B) D (C - B)^T v = ((C - B)^T v)^T D ((C - B)^T v) \geq 0.$$

Consequently,

$$\mathrm{Var}(\widehat{\beta}_{WLS}) - \mathrm{Var}(\widehat{\beta}_{OLS}) = -\sigma^2 (C - B) D (C - B)^T \preceq 0.$$

This is precisely the negative-semidefinite comparison requested in the question.

Final answer

Use weighted least squares with weights inversely proportional to the known error variances:

$$\boxed{\widehat{\beta}_{WLS} = (X^T D^{-1} X)^{-1} X^T D^{-1} Y, \qquad D = \mathrm{Diag}(t, t^2, \ldots, t^n).}$$

Its covariance is $\sigma^2 (X^T D^{-1} X)^{-1}$ and

$$\boxed{\mathrm{Var}(\widehat{\beta}_{WLS}) - \mathrm{Var}(\widehat{\beta}_{OLS}) = -\sigma^2 (C - B) D (C - B)^T \preceq 0.}$$

Thus WLS is unbiased and has covariance no larger than OLS under the stated known heteroskedastic covariance structure.

4 Question 4: Expected Mahalanobis distances and hat-matrix entries

Question (as assigned)

4. Let $x^{(1)}, \ldots, x^{(n)}$ be n observations on p-covariates, i.e. $x^{(i)} \in \mathbb{R}^p$, and define

$$\bar{x} = n^{-1} \sum_{i=1}^{n} x^{(i)}, \qquad S_X = n^{-1} \sum_{i=1}^{n} (x^{(i)} - \bar{x})(x^{(i)} - \bar{x})^T.$$

Consider the Mahalanobis distances

$$MD_i = \{(x^{(i)} - \bar{x})^T S_X^{-1} (x^{(i)} - \bar{x})\}^{1/2},$$

$$MD_{i,j} = \{(x^{(i)} - x^{(j)})^T S_X^{-1} (x^{(i)} - x^{(j)})\}^{1/2}.$$

Suppose $x^{(i)}$'s are i.i.d. samples from a p-variate distribution with finite variance-covariance matrix.

(a) Show that $\mathbb{E}(MD_i^2) = p$.

(b) Show that $\mathbb{E}(MD_{i,j}^2) = 2np/(n-1)$.

(c) Define $Z = [(x^{(1)})^T; \ldots; (x^{(n)})^T]$, $X = [\mathbf{1}\ Z]$ and $P(X) = X(X^TX)^{-1}X^T = ((p_{i,j}))$. Using (i) and (ii), derive the expectations of $p_{i,i}$ and $p_{i,j}$ for $i \neq j$.

Required theory from the supplied lecture notes

The supplied notes define Mahalanobis distance from the sample center by

$$MD_i^2 = (x^{(i)} - \bar{x})^T S_X^{-1}(x^{(i)} - \bar{x}),$$

and give the hat-matrix identity for a model with an intercept:

$$p_{ii} = \frac{1}{n}\left\{1 + (x^{(i)} - \bar{x})^T S_X^{-1}(x^{(i)} - \bar{x})\right\} = \frac{1 + MD_i^2}{n}.$$

For a projection matrix, $P^T = P$, $P^2 = P$, and, when the intercept is included, $P\mathbf{1} = \mathbf{1}$. Hence every row of P sums to one. The trace of an idempotent projection is its rank.

Regularity note. Since the formula uses S_X^{-1}, we assume the sample covariance matrix is nonsingular (and hence $n > p$). The assignment's inverse notation implicitly requires this condition.

Part (a): Show that $\mathbb{E}(MD_i^2) = p$

Step 1: Sum the squared distances over all sample points

For a fixed realized sample, use the identity $a^TBa = \operatorname{tr}(Baa^T)$:

$$\begin{aligned}\sum_{i=1}^n MD_i^2 &= \sum_{i=1}^n \operatorname{tr}\left[S_X^{-1}(x^{(i)} - \bar{x})(x^{(i)} - \bar{x})^T\right] \\ &= \operatorname{tr}\left[S_X^{-1}\sum_{i=1}^n (x^{(i)} - \bar{x})(x^{(i)} - \bar{x})^T\right].\end{aligned}$$

The matrix inside the sum is exactly nS_X by the definition of S_X.

Step 2: Simplify the trace

Therefore, sample by sample,

$$\sum_{i=1}^n MD_i^2 = \operatorname{tr}(S_X^{-1}nS_X) = n\operatorname{tr}(I_p) = np.$$

This identity does not yet use expectation.

Step 3: Use exchangeability

Because the observations are i.i.d., the random variables $MD_1^2, \ldots, MD_n^2$ have the same expectation. Taking expectations of the sum gives

$$n\mathbb{E}(MD_i^2) = np.$$

Dividing by n yields the required result.

Final answer

$$\boxed{\mathbb{E}(MD_i^2) = p.}$$

The key is that the sum of all squared Mahalanobis distances from the sample center is exactly np for every nonsingular realized sample.

Part (b): Show that $\mathbb{E}(MD_{i,j}^2) = 2np/(n-1)$

Step 1: Establish a pairwise-difference identity

For any vectors $x^{(1)}, \ldots, x^{(n)}$,

$$\sum_{1 \le i < j \le n} (x^{(i)} - x^{(j)})(x^{(i)} - x^{(j)})^T = n \sum_{i=1}^{n} (x^{(i)} - \bar{x})(x^{(i)} - \bar{x})^T.$$

To see why, expand the left side. Each $x^{(i)}x^{(i)T}$ appears $n-1$ times, while the cross-products combine to $-\sum_{i \neq j} x^{(i)}x^{(j)T}$. Rewriting the result in terms of $\sum_i x^{(i)}x^{(i)T}$ and $n\bar{x}\bar{x}^T$ gives the right side.

Step 2: Convert the matrix identity into Mahalanobis distances

Premultiply by S_X^{-1} and take traces. The result is

$$\begin{aligned} \sum_{i<j} MD_{i,j}^2 &= n \operatorname{tr}\left[S_X^{-1} \sum_i (x^{(i)} - \bar{x})(x^{(i)} - \bar{x})^T \right] \\ &= n \operatorname{tr}(S_X^{-1} n S_X) = n^2 p. \end{aligned}$$

Step 3: Use exchangeability over the pairs

There are $\binom{n}{2} = n(n-1)/2$ unordered pairs. The i.i.d. assumption makes the expectation of $MD_{i,j}^2$ the same for every pair. Thus

$$\frac{n(n-1)}{2} \mathbb{E}(MD_{i,j}^2) = n^2 p.$$

Solving gives

$$\mathbb{E}(MD_{i,j}^2) = \frac{2np}{n-1}.$$

Final answer

$$\boxed{\mathbb{E}(MD_{i,j}^2) = \frac{2np}{n-1}, \qquad i \neq j.}$$

Part (c): Expectations of the diagonal and off-diagonal hat-matrix entries

Step 1: Use the hat-matrix Mahalanobis identity

The design matrix is $X = [\mathbf{1}\ Z]$, so it contains an intercept and the p covariate columns. The lecture-note identity gives

$$p_{ii} = \frac{1 + MD_i^2}{n}.$$

Taking expectation and using part (a),

$$\boxed{\mathbb{E}(p_{ii}) = \frac{1+p}{n}.}$$

Step 2: Use the row-sum property

Since $P_X\mathbf{1} = \mathbf{1}$, its ith component says

$$\sum_{j=1}^{n} p_{ij} = 1.$$

Taking expectation,

$$\mathbb{E}(p_{ii}) + \sum_{j\neq i} \mathbb{E}(p_{ij}) = 1.$$

By exchangeability of the observations, all off-diagonal expectations are equal. Write their common value as c. Then

$$\frac{p+1}{n} + (n-1)c = 1.$$

Step 3: Solve for the off-diagonal expectation

$$(n-1)c = \frac{n-p-1}{n}, \qquad c = \frac{n-p-1}{n(n-1)}.$$

Final answer

$$\boxed{\mathbb{E}(p_{ii}) = \frac{p+1}{n}, \qquad \mathbb{E}(p_{ij}) = \frac{n-p-1}{n(n-1)} \quad (i \neq j).}$$

The diagonal expectation comes from Mahalanobis distance; the off-diagonal expectation follows from the fact that each hat-matrix row sums to one.

5 Question 5: Extra sum of squares and the covariance of the added coefficients

Question (as assigned)

5. Let Y be the vector of n-observations on a response variable Y, $\{x_1, \ldots, x_p, z_1, \ldots, z_q\}$ be the corresponding vectors for $p+q$ covariates, and $X = [\mathbf{1}\ x_1\ \cdots\ x_p]$, $Z = [z_1\ \cdots\ z_q]$ and $W = [X : Z]$. Consider two models:

$$Y = X\beta + Z\gamma + \epsilon^\star = W\delta + \epsilon^\star; \tag{3}$$

$$Y = X\beta + \epsilon. \tag{4}$$

where $\mathbb{E}(\epsilon) = \mathbb{E}(\epsilon^\star) = 0$, and $\text{var}(\epsilon) = \text{var}(\epsilon^\star) = \sigma^2 I$. Let $\widehat{\gamma}_{OLS}$ be the OLS estimator of γ based on model (3). Then show that

$$Y^T\{P(X) - P(W)\}Y = \sigma^2\widehat{\gamma}_{OLS}^T\{\text{Var}(\widehat{\gamma}_{OLS})\}^{-1}\widehat{\gamma}_{OLS}.$$

Sign correction required by the printed statement

The left side is printed as $Y^T\{P(X) - P(W)\}Y$. Since $\mathcal{C}(X) \subseteq \mathcal{C}(W)$, the difference $P(W) - P(X)$ is a positive-semidefinite projection, whereas $P(X) - P(W)$ is negative semidefinite. The right side in the printed equality is nonnegative. Therefore the equality as printed has the wrong sign. The mathematically correct identity is

$$Y^T\{P(W) - P(X)\}Y = \sigma^2\widehat{\gamma}_{OLS}^T\{\text{Var}(\widehat{\gamma}_{OLS})\}^{-1}\widehat{\gamma}_{OLS}.$$

We retain the assignment's printed version in the question above and prove the corrected identity below. Equivalently, the printed left side equals the negative of the right side.

Required theory from the supplied lecture notes

For nested regression spaces $\mathcal{C}(X) \subseteq \mathcal{C}(W)$, the projection onto the larger space contains the projection onto the smaller space. Residualizing the columns of Z against X gives

$$M_X Z = (I - P_X)Z, \qquad H = Z^T M_X Z.$$

If W has full column rank, H is invertible and the additional part of the projection is

$$P_W - P_X = M_X Z(Z^T M_X Z)^{-1} Z^T M_X.$$

The partitioned least-squares equations (equivalently, the partial-regression or Frisch–Waugh–Lovell calculation) give

$$\widehat{\gamma}_{OLS} = (Z^T M_X Z)^{-1} Z^T M_X Y, \qquad \mathrm{Var}(\widehat{\gamma}_{OLS}) = \sigma^2 (Z^T M_X Z)^{-1}.$$

These formulas are also consistent with the block-matrix inversion and orthogonal-projection theory in the supplied notes.

Purpose of the proof. The additional sum of squares explained by Z after X has been fitted is a quadratic form in the estimated extra coefficients. We make this statement exact.

Step 1: Residualize the added regressors against the original design

Let $M_X = I - P_X$ and put $Z_\perp = M_X Z$. Each column of $Z_\perp$ is orthogonal to $\mathcal{C}(X)$. Moreover,

$$\mathcal{C}(W) = \mathcal{C}(X) \oplus \mathcal{C}(Z_\perp),$$

where the sum is orthogonal. Therefore the projection onto $\mathcal{C}(W)$ is the sum of the projections onto these two orthogonal spaces.

Step 2: Write the difference between the two projection matrices

The projection onto the columns of $Z_\perp$ is

$$P_{Z_\perp} = Z_\perp (Z_\perp^T Z_\perp)^{-1} Z_\perp^T.$$

Since $Z_\perp^T Z_\perp = Z^T M_X Z$ and $Z_\perp^T = Z^T M_X$,

$$\begin{aligned} P_W - P_X &= P_{Z_\perp} \\ &= M_X Z(Z^T M_X Z)^{-1} Z^T M_X. \end{aligned}$$

Step 3: Express the additional sum of squares

Premultiply and postmultiply by Y^T and Y:

$$Y^T(P_W - P_X)Y = Y^T M_X Z(Z^T M_X Z)^{-1} Z^T M_X Y.$$

By the partial-regression formula,

$$\widehat{\gamma}_{OLS} = (Z^T M_X Z)^{-1} Z^T M_X Y.$$

Thus $Z^T M_X Y = (Z^T M_X Z)\widehat{\gamma}_{OLS}$, and substitution gives

$$Y^T(P_W - P_X)Y = \widehat{\gamma}_{OLS}^T (Z^T M_X Z)\widehat{\gamma}_{OLS}.$$

Step 4: Replace the middle matrix by the covariance matrix

Under model (3),

$$\begin{aligned}\mathrm{Var}(\widehat{\gamma}_{OLS}) &= H^{-1}Z^T M_X \,\mathrm{Var}(Y) M_X Z H^{-1} \\ &= H^{-1}Z^T M_X(\sigma^2 I) M_X Z H^{-1} \\ &= \sigma^2 H^{-1}Z^T M_X Z H^{-1} \\ &= \sigma^2 H^{-1},\end{aligned}$$

where we used $M_X^2 = M_X$. Consequently,

$$\{\mathrm{Var}(\widehat{\gamma}_{OLS})\}^{-1} = \frac{1}{\sigma^2} H.$$

Therefore

$$\sigma^2 \widehat{\gamma}_{OLS}^T \{\mathrm{Var}(\widehat{\gamma}_{OLS})\}^{-1} \widehat{\gamma}_{OLS} = \widehat{\gamma}_{OLS}^T H \widehat{\gamma}_{OLS},$$

which is exactly the expression in Step 3.

Final answer

The correct identity is

$$\boxed{Y^T\{P(W) - P(X)\}Y = \sigma^2 \widehat{\gamma}_{OLS}^T \{\mathrm{Var}(\widehat{\gamma}_{OLS})\}^{-1} \widehat{\gamma}_{OLS}.}$$

The assignment's printed order $P(X) - P(W)$ gives the negative of this quantity. The result is the quadratic-form expression for the extra sum of squares due to adding Z after accounting for X.

6 Question 6: Deleted residuals and the deleted-sample hat matrix

Question (as assigned)

6. Consider fitting a full-rank multiple regression model $y = X\beta + \epsilon$ by the least-squares method. Let $P(X) = ((p_{ij}))$ be the orthogonal projection matrix onto $\mathcal{C}(X)$ and let $P_{(i)} = ((p_{jk(i)}))$ be the same when the ith observation was removed. Let x_i be the ith row of X as a column vector, e_i be the ith residual from the full model fit, and $\widehat{\beta}_{(i)}$ be the ith deleted OLS estimate of β.

(a) Show that $e_{i(i)} = e_i/(1 - p_{ii})$, where $e_{i(i)} = y_i - x_i^T \widehat{\beta}_{(i)}$.

(b) Prove that $p_{jk(i)} = p_{jk} + p_{ik}p_{ij}/(1 - p_{ii})$. [You may assume $\max\{j, k\} < i$.]

(c) In case of no-intercept model, show that $-0.5 \le p_{ij} \le 0.5$, $i \ne j$.

Required theory from the supplied lecture notes

The supplied leave-one-out notes state the coefficient-deletion identity

$$\widehat{\beta}_{OLS} - \widehat{\beta}_{OLS,(-i)} = (X^T X)^{-1} x_i \frac{e_i}{1 - p_{ii}}.$$

The hat matrix is symmetric and idempotent:

$$P^T = P, \qquad P^2 = P.$$

The Sherman–Morrison rank-one inverse formula is

$$(A - uu^T)^{-1} = A^{-1} + \frac{A^{-1}uu^T A^{-1}}{1 - u^T A^{-1} u},$$

when the displayed inverse exists. Both P and $I - P$ are symmetric idempotent matrices; therefore both are positive semidefinite.

Part (a): Deleted residual identity

Step 1: Start from the deleted-estimator formula

The lecture-note identity gives

$$\widehat{\beta}_{(i)} = \widehat{\beta} - (X^T X)^{-1} x_i \frac{e_i}{1 - p_{ii}}.$$

Step 2: Substitute into the definition of the deleted residual

By definition,

$$\begin{aligned} e_{i(i)} &= y_i - x_i^T \widehat{\beta}_{(i)} \\ &= y_i - x_i^T \left[\widehat{\beta} - (X^T X)^{-1} x_i \frac{e_i}{1 - p_{ii}} \right] \\ &= \underbrace{(y_i - x_i^T \widehat{\beta})}_{e_i} + x_i^T (X^T X)^{-1} x_i \frac{e_i}{1 - p_{ii}}. \end{aligned}$$

Step 3: Recognize the leverage value

Since $p_{ii} = x_i^T (X^T X)^{-1} x_i$,

$$\begin{aligned} e_{i(i)} &= e_i + p_{ii} \frac{e_i}{1 - p_{ii}} \\ &= e_i \left(\frac{1 - p_{ii} + p_{ii}}{1 - p_{ii}} \right) = \frac{e_i}{1 - p_{ii}}. \end{aligned}$$

Final answer

$$\boxed{e_{i(i)} = \frac{e_i}{1 - p_{ii}}.}$$

The deleted residual is the full-model residual inflated by the factor $(1 - p_{ii})^{-1}$.

Part (b): Formula for the deleted hat-matrix entries

Step 1: Write the cross-product matrix after deleting row i

Let $A = X^T X$. Removing observation i subtracts its outer product, so

$$X_{(-i)}^T X_{(-i)} = A - x_i x_i^T.$$

Step 2: Apply Sherman–Morrison

With $u = x_i$,

$$(A - x_i x_i^T)^{-1} = A^{-1} + \frac{A^{-1} x_i x_i^T A^{-1}}{1 - x_i^T A^{-1} x_i}.$$

The denominator is $1 - p_{ii}$.

Step 3: Evaluate a generic deleted hat-matrix entry

For rows j, k that remain in the sample,

$$\begin{aligned} p_{jk(i)} &= x_j^T (X_{(-i)}^T X_{(-i)})^{-1} x_k \\ &= x_j^T A^{-1} x_k + \frac{x_j^T A^{-1} x_i \, x_i^T A^{-1} x_k}{1 - p_{ii}} \\ &= p_{jk} + \frac{p_{ji} p_{ik}}{1 - p_{ii}}. \end{aligned}$$

The hat matrix is symmetric, so $p_{ji} = p_{ij}$. Hence

$$p_{jk(i)} = p_{jk} + \frac{p_{ik}p_{ij}}{1 - p_{ii}}.$$

Final answer

$$\boxed{p_{jk(i)} = p_{jk} + \frac{p_{ik}p_{ij}}{1 - p_{ii}}.}$$

This is a rank-one update of the original hat matrix after the ith observation is removed.

Part (c): Bound the off-diagonal hat-matrix entries

Step 1: Use positive semidefiniteness of P

Because P is an orthogonal projection, it is positive semidefinite. Its 2×2 principal submatrix indexed by i, j is positive semidefinite, so its determinant is nonnegative:

$$\det \begin{pmatrix} p_{ii} & p_{ij} \\ p_{ij} & p_{jj} \end{pmatrix} = p_{ii}p_{jj} - p_{ij}^2 \geq 0.$$

Thus

$$p_{ij}^2 \leq p_{ii}p_{jj}.$$

Step 2: Use positive semidefiniteness of $I - P$

The matrix $I - P$ is also an orthogonal projection. Its corresponding off-diagonal entry is $-p_{ij}$ and its diagonal entries are $1 - p_{ii}$ and $1 - p_{jj}$. Therefore

$$p_{ij}^2 \leq (1 - p_{ii})(1 - p_{jj}).$$

Step 3: Combine the two inequalities

Multiplying the two upper bounds gives

$$p_{ij}^4 \leq p_{ii}p_{jj}(1 - p_{ii})(1 - p_{jj}) = \{p_{ii}(1 - p_{ii})\}\{p_{jj}(1 - p_{jj})\}.$$

For every $a \in [0, 1]$, $a(1 - a) \leq 1/4$. Since projection diagonals lie in $[0, 1]$,

$$p_{ij}^4 \leq \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}.$$

Taking fourth roots gives $|p_{ij}| \leq 1/2$.

Final answer

$$\boxed{-\frac{1}{2} \leq p_{ij} \leq \frac{1}{2}, \qquad i \neq j.}$$

In fact, the proof uses only that P is an orthogonal projection; the no-intercept condition is not needed for this bound.

7 Question 7: Increase in RSS after adding one observation

Question (as assigned)

7. Consider fitting a regression model with regression matrix X, response vector Y and unknown parameter vector β. Let $\widehat{\beta}$ be the OLS estimate of β. Suppose that a new observation (x_{n+1}, Y_{n+1}) is added. Show that the residual sum of squares is increased by an amount

$$\frac{e_{n+1}^2}{1 + x_{n+1}^T (X^T X)^{-1} x_{n+1}}, \qquad e_{n+1} = Y_{n+1} - x_{n+1}^T \widehat{\beta}.$$

Required theory from the supplied lecture notes

OLS minimizes $\|Y - X\beta\|^2$. At the OLS estimate, the residual vector $e = Y - X\widehat{\beta}$ is orthogonal to every column of X, so $X^T e = 0$. Therefore, if the coefficient vector is perturbed by d,

$$\|Y - X(\widehat{\beta} + d)\|^2 = \mathrm{RSS} + d^T X^T X d.$$

The Sherman–Morrison identity for a rank-one addition is

$$(A + uu^T)^{-1} = A^{-1} - \frac{A^{-1} u u^T A^{-1}}{1 + u^T A^{-1} u}.$$

Plan. We express the new residual sum of squares as a function of the change d in the old coefficient estimate, minimize that one quadratic expression, and identify the resulting increment.

Step 1: Parameterize a candidate fit by its change from the old estimate

Let $x = x_{n+1}$, $y = Y_{n+1}$ and $e = y - x^T \widehat{\beta}$. Write a candidate coefficient vector as $\widehat{\beta} + d$. For the original n observations,

$$\begin{aligned} \|Y - X(\widehat{\beta} + d)\|^2 &= \|e_{old} - Xd\|^2 \\ &= \|e_{old}\|^2 - 2e_{old}^T X d + d^T X^T X d \\ &= \mathrm{RSS} + d^T X^T X d, \end{aligned}$$

because $X^T e_{old} = 0$.

Step 2: Add the contribution of the new observation

The new observation has residual $y - x^T(\widehat{\beta} + d) = e - x^T d$. Thus the enlarged RSS, for a given d, is

$$\mathrm{RSS}_{new}(d) = \mathrm{RSS} + d^T A d + (e - x^T d)^2, \qquad A = X^T X.$$

Only the last two terms need to be minimized.

Step 3: Find the minimizing coefficient adjustment

Expand and differentiate with respect to d:

$$d^T A d + (e - x^T d)^2 = d^T A d + e^2 - 2e x^T d + d^T x x^T d.$$

The gradient is

$$2Ad - 2ex + 2xx^T d = 0,$$

so

$$(A + xx^T)d = xe, \qquad d = (A + xx^T)^{-1} xe.$$

Step 4: Evaluate the minimum using Sherman–Morrison

Put $h = x^T A^{-1} x$. Sherman–Morrison gives

$$(A + xx^T)^{-1} = A^{-1} - \frac{A^{-1} x x^T A^{-1}}{1 + h}.$$

Consequently,

$$x^T (A + xx^T)^{-1} x = h - \frac{h^2}{1 + h} = \frac{h}{1 + h}.$$

The minimum of $d^T A d + (e - x^T d)^2$ can be written as

$$e^2 - e^2 x^T (A + xx^T)^{-1} x = e^2 \left(1 - \frac{h}{1 + h}\right) = \frac{e^2}{1 + h}.$$

Since $h = x^T (X^T X)^{-1} x$, the increase over the old RSS is the required amount.

Final answer

$$\boxed{\mathrm{RSS}_{new} - \mathrm{RSS}_{old} = \frac{e_{n+1}^2}{1 + x_{n+1}^T (X^T X)^{-1} x_{n+1}}.}$$

The numerator is the squared prediction error of the new point under the old fit; the denominator accounts for the ability of the refitted coefficients to adjust to that point.

8 Question 8: An unbiased estimator under equicorrelated errors

Question (as assigned)

8. Consider the model in (1), except that $\epsilon \sim N(0, \sigma^2 V)$ where

$$V = (1 - \rho) I + \rho \mathbf{1}\mathbf{1}^T,$$

for some $\rho > 0$. Find an unbiased estimator of σ^2 when ρ is known. Express the unbiased estimator in terms of RSS, $(\bar{Y}, \bar{x}_1, \ldots, \bar{x}_p)$, (n, p) and ρ.

Required theory from the supplied lecture notes

For a regression model with mean $X\beta$ and residual-maker $M_X = I - P_X$,

$$\mathrm{RSS} = Y^T M_X Y.$$

Because $M_X X = 0$, the mean contribution disappears from the residual sum of squares. If $\mathrm{Var}(\epsilon) = \sigma^2 V$, then

$$\mathbb{E}(\mathrm{RSS}) = \mathrm{tr}\{M_X \, \mathrm{Var}(\epsilon)\} = \sigma^2 \, \mathrm{tr}(M_X V).$$

The design contains an intercept, so $M_X \mathbf{1} = 0$. Also $\mathrm{tr}(M_X) = n - p - 1$.

A parameter restriction. Since V must be a covariance matrix, its eigenvalues must be nonnegative. The eigenvalue along $\mathbf{1}$ is $1 + (n - 1)\rho$, and the $n - 1$ eigenvalues orthogonal to $\mathbf{1}$ are $1 - \rho$. With the stated $\rho > 0$, positive definiteness requires $0 < \rho < 1$.

Step 1: Write RSS as a quadratic form in the errors

Let $M_X = I - P_X$. Since $M_X X = 0$,

$$\mathrm{RSS} = Y^T M_X Y = (X\beta + \epsilon)^T M_X (X\beta + \epsilon) = \epsilon^T M_X \epsilon.$$

Step 2: Take the expectation

Using $\mathbb{E}(\epsilon) = 0$ and $\mathrm{Var}(\epsilon) = \sigma^2 V$,

$$\mathbb{E}(\mathrm{RSS}) = \sigma^2 \operatorname{tr}(M_X V).$$

Insert the equicorrelation form of V:

$$\begin{aligned}\operatorname{tr}(M_X V) &= \operatorname{tr}\{M_X[(1-\rho)I + \rho \mathbf{1}\mathbf{1}^T]\} \\ &= (1-\rho)\operatorname{tr}(M_X) + \rho \operatorname{tr}(M_X \mathbf{1}\mathbf{1}^T).\end{aligned}$$

Step 3: Use the intercept to eliminate the common-correlation term

Because $M_X \mathbf{1} = 0$,

$$M_X \mathbf{1}\mathbf{1}^T = 0, \qquad \operatorname{tr}(M_X \mathbf{1}\mathbf{1}^T) = 0.$$

Also $\operatorname{tr}(M_X) = n - p - 1$. Hence

$$\mathbb{E}(\mathrm{RSS}) = \sigma^2(1-\rho)(n-p-1).$$

Step 4: Construct an unbiased estimator

Divide RSS by its expectation multiplier:

$$\widehat{\sigma}^2 = \frac{\mathrm{RSS}}{(1-\rho)(n-p-1)}.$$

Indeed, $\mathbb{E}(\widehat{\sigma}^2) = \sigma^2$.

Step 5: Show explicitly how the sample means enter the RSS

Because the model has an intercept, OLS can be written using centered response and centered predictors. Define

$$Y_c = Y - \bar{Y}\mathbf{1}, \qquad Z_c = [\, x_1 - \bar{x}_1 \mathbf{1} \;\cdots\; x_p - \bar{x}_p \mathbf{1} \,].$$

Then

$$M_X = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T - P_{Z_c}, \qquad P_{Z_c} = Z_c (Z_c^T Z_c)^{-1} Z_c^T,$$

and therefore

$$\mathrm{RSS} = Y_c^T (I - P_{Z_c}) Y_c.$$

Thus the requested expression in terms of the sample means is

$$\boxed{\widehat{\sigma}^2 = \frac{(Y - \bar{Y}\mathbf{1})^T \left[I - Z_c (Z_c^T Z_c)^{-1} Z_c^T \right] (Y - \bar{Y}\mathbf{1})}{(1-\rho)(n-p-1)}}.$$

The means enter through centering; no additional separate correction involving $\bar{Y}$ or the $\bar{x}_j$ is needed because the common component $\rho \mathbf{1}\mathbf{1}^T$ is annihilated by residualization against the intercept.

Final answer

$$\boxed{\widehat{\sigma}^2_{unbiased} = \frac{\mathrm{RSS}}{(1-\rho)(n-p-1)}}.$$

Equivalently, compute RSS after centering Y and all predictors at their sample means, as in Step 5, and divide by $(1-\rho)(n-p-1)$. The intercept is essential: it makes $M_X \mathbf{1} = 0$ and removes the equicorrelated common component from RSS.

9 Question 9: Covariance matrix and distribution of AR(1) errors

Question (as assigned)

9. In a linear regression setup, let Y be the response and X be a predictor of Y. Based on n observations, consider the model

$$Y_t = \alpha + \beta X_t + \epsilon_t,$$

where

$$\epsilon_1 \sim N\left(0, \frac{\sigma^2}{1-\rho^2}\right), \qquad \epsilon_t = \rho\epsilon_{t-1} + u_t, \quad t = 2, \ldots, T,$$

and

$$(u_2, \ldots, u_T)^T \sim N_{T-1}(0, \sigma^2 I), \qquad \epsilon_1 \perp\!\!\!\perp (u_2, \ldots, u_T)^T.$$

Find the variance-covariance matrix and hence the distribution of ϵ.

Required theory from the supplied lecture notes

The supplied notes define the stationary AR(1) error structure by

$$\epsilon_t = \rho\epsilon_{t-1} + u_t, \qquad u_t \text{ independent with variance } \sigma^2,$$

with the initial variance chosen as $\sigma^2/(1-\rho^2)$. They derive

$$\mathrm{Var}(\epsilon_t) = \frac{\sigma^2}{1-\rho^2}, \qquad \mathrm{Cov}(\epsilon_t, \epsilon_{t+h}) = \rho^h \frac{\sigma^2}{1-\rho^2}.$$

A linear transformation of jointly normal independent variables is multivariate normal.

Step 1: Iterate the AR(1) recursion

Repeated substitution gives

$$\begin{aligned}
\epsilon_1 &= \epsilon_1, \\
\epsilon_2 &= \rho\epsilon_1 + u_2, \\
\epsilon_3 &= \rho^2\epsilon_1 + \rho u_2 + u_3, \\
\epsilon_4 &= \rho^3\epsilon_1 + \rho^2 u_2 + \rho u_3 + u_4,
\end{aligned}$$

and, in general,

$$\epsilon_t = \rho^{t-1}\epsilon_1 + \sum_{s=2}^{t} \rho^{t-s} u_s.$$

Step 2: Find the variance of each error

The initial variance is $\mathrm{Var}(\epsilon_1) = \sigma^2/(1-\rho^2)$. Since u_t is independent of ϵ_{t-1},

$$\mathrm{Var}(\epsilon_t) = \rho^2 \,\mathrm{Var}(\epsilon_{t-1}) + \sigma^2.$$

If $\mathrm{Var}(\epsilon_{t-1}) = \sigma^2/(1-\rho^2)$, then

$$\mathrm{Var}(\epsilon_t) = \frac{\rho^2\sigma^2}{1-\rho^2} + \sigma^2 = \frac{\sigma^2}{1-\rho^2}.$$

Since this is true for $t = 1$, induction gives it for every t.

Step 3: Find the covariance at two different times

Take $s < t$. From the recursion,

$$\epsilon_t = \rho^{t-s}\epsilon_s + \sum_{k=s+1}^{t} \rho^{t-k} u_k.$$

The future innovations $u_{s+1}, \ldots, u_t$ are independent of ϵ_s, so

$$\begin{aligned}\operatorname{Cov}(\epsilon_s, \epsilon_t) &= \rho^{t-s} \operatorname{Var}(\epsilon_s) \\ &= \rho^{t-s} \frac{\sigma^2}{1-\rho^2}.\end{aligned}$$

By symmetry, for any s, t,

$$\operatorname{Cov}(\epsilon_s, \epsilon_t) = \frac{\sigma^2}{1-\rho^2} \rho^{|t-s|}.$$

Step 4: Assemble the covariance matrix

The diagonal entries are $\sigma^2/(1-\rho^2)$, the first off-diagonals are $\rho\sigma^2/(1-\rho^2)$, the second off-diagonals are $\rho^2\sigma^2/(1-\rho^2)$, and so on. Therefore

$$\Sigma_\epsilon = \frac{\sigma^2}{1-\rho^2} \begin{pmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{T-1} \\ \rho & 1 & \rho & \cdots & \rho^{T-2} \\ \rho^2 & \rho & 1 & \cdots & \rho^{T-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{T-1} & \rho^{T-2} & \rho^{T-3} & \cdots & 1 \end{pmatrix}.$$

Step 5: State the joint distribution

The vector $(\epsilon_1, u_2, \ldots, u_T)^T$ has independent normal components by the assumptions. The recursion expresses $\epsilon = (\epsilon_1, \ldots, \epsilon_T)^T$ as a linear transformation of that vector. Hence ϵ is multivariate normal with mean zero and covariance matrix just derived:

$$\epsilon \sim N_T(0, \Sigma_\epsilon).$$

The stationary initialization requires $|\rho| < 1$.

Final answer

$$\boxed{\operatorname{Cov}(\epsilon_s, \epsilon_t) = \frac{\sigma^2}{1-\rho^2} \rho^{|t-s|}}$$

and

$$\boxed{\epsilon \sim N_T\left(0, \frac{\sigma^2}{1-\rho^2} \begin{pmatrix} 1 & \rho & \cdots & \rho^{T-1} \\ \rho & 1 & \cdots & \rho^{T-2} \\ \vdots & \vdots & \ddots & \vdots \\ \rho^{T-1} & \rho^{T-2} & \cdots & 1 \end{pmatrix}\right).}$$

10 Question 10: Effect of scaling on VIF and condition number

Question (as assigned)

10. Consider the model in (1). Recall the variance inflation factor VIF_j, $j = 1, \ldots, p$, and the condition number $\kappa(X)$. Suppose the covariates are scaled by their respective standard deviations. Will the value of VIF_j change after this scaling? Will the value of $\kappa(X)$ change after this scaling? Justify your answers.

Required theory from the supplied lecture notes

The supplied notes define

$$VIF_j = \frac{\|x_j\|^2}{x_j^T\{I - P(X_{(-j)})\}x_j},$$

where $X_{(-j)}$ contains the other regressors (and the intercept, when present). A large VIF signals near collinearity.
The notes define the condition number by

$$\kappa(X) = \sqrt{\frac{\lambda_{\max}(X^TX)}{\lambda_{\min}(X^TX)}} = \frac{\sigma_{\max}(X)}{\sigma_{\min}(X)}.$$

It measures the imbalance between the largest and smallest singular-value directions of the design.

Step 1: Represent the scaling as a column transformation

Let $s_j > 0$ be the standard deviation used to scale column x_j. Define the diagonal scaling matrix

$$D_s = \text{Diag}(1, s_1, \ldots, s_p), \qquad X^* = XD_s^{-1},$$

where the intercept is left unchanged. Thus $x_j^* = x_j/s_j$.

Step 2: Show that VIF is invariant

Scaling the other predictor columns does not change their column space. Therefore

$$P(X^*_{(-j)}) = P(X_{(-j)}).$$

The numerator of the scaled VIF is

$$\|x_j^*\|^2 = \frac{1}{s_j^2}\|x_j\|^2.$$

The denominator is

$$(x_j^*)^T\{I - P(X^*_{(-j)})\}x_j^* = \frac{1}{s_j^2}x_j^T\{I - P(X_{(-j)})\}x_j.$$

The numerator and denominator are multiplied by exactly the same factor s_j^{-2}. Hence

$$VIF_j^* = VIF_j.$$

Step 3: Check what happens to the condition number

The scaled Gram matrix is

$$(X^*)^TX^* = D_s^{-1}X^TXD_s^{-1}.$$

This is generally not a scalar multiple of X^TX. Its eigenvalues therefore need not preserve their ratio. A simple illustration is a Gram matrix

$$X^TX = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix},$$

whose condition number (under the notes' square-root convention) is $\sqrt{4/1} = 2$. Scaling the second column by 2 changes the Gram matrix to the identity, whose condition number is 1. This demonstrates that column scaling can change $\kappa(X)$.

It is possible for the condition number to remain the same in special cases, but unlike VIF it is not invariant to arbitrary columnwise rescaling.

Final answer

$\boxed{VIF_j \text{ does not change under nonzero scaling of the covariates.}}$

$\boxed{\kappa(X) \text{ generally changes under columnwise scaling; it is not scale-invariant.}}$

The first result follows because VIF is a ratio in which the same squared scale factor cancels. The second follows because the eigenvalue ratio of $D_s^{-1}X^TXD_s^{-1}$ is generally different from that of X^TX.

11 Question 11: Projection representation and Beta law of a standardized residual

Question (as assigned)

11. Consider the regression model in (1), and let $P = P(X) = X(X^TX)^{-1}X^T$ be the hat matrix.

(a) Express e_i in the form $e_i = c_i^T(I_n - P)\epsilon$ for a suitable vector c_i.

(b) Show that $(n-p-1)^{-1}r_i^2$ can be written as

$$\frac{Z^TQZ}{Z^T(I_n-P)Z}, \qquad Q = (1-p_{ii})^{-1}(I_n-P)c_ic_i^T(I_n-P),$$

where r_i is the standardized residual and $Z \sim N_n(0, I_n)$.

(c) Prove that Q is a projection matrix (i.e. show that $Q^2 = Q$ and $Q^T = Q$).

(d) Show that $I_n - P - Q$ is a projection matrix and prove that Z^TQZ and $Z^T(I_n-P-Q)Z$ are independent. Hence prove that $(n-p-1)^{-1}r_i^2$ has a Beta$\{1/2, \frac{1}{2}(n-p)-1\}$ distribution.

Required theory from the supplied lecture notes

The lecture notes define the residual vector and internally standardized residual by

$$e = (I - P_X)Y, \qquad r_i = \frac{e_i}{\widehat{\sigma}_{UE}\sqrt{1-p_{ii}}}, \qquad \widehat{\sigma}^2_{UE} = \frac{\text{RSS}}{n-p-1}.$$

Under the normal linear model,

$$\frac{\text{RSS}}{\sigma^2} \sim \chi^2_{n-p-1}.$$

For a standard normal vector Z, a symmetric idempotent matrix A of rank r gives $Z^TAZ \sim \chi^2_r$. If two such projection matrices have orthogonal ranges, their quadratic forms are independent. Finally, if $U \sim \chi^2_a$ and $V \sim \chi^2_b$ independently, then

$$\frac{U}{U+V} \sim \text{Beta}\left(\frac{a}{2}, \frac{b}{2}\right).$$

Notation for the proof. Write $M = I_n - P$, and let c_i be the ith coordinate vector (one in coordinate i, zero elsewhere). This distinguishes the coordinate vector from the scalar residual e_i.

Part (a): Express e_i as a linear form in the errors

Step 1: Write the residual vector using the projection matrix

Since $Y = X\beta + \epsilon$ and $MX = 0$,

$$e = MY = M(X\beta + \epsilon) = M\epsilon.$$

Step 2: Select the required coordinate

The ith residual is the ith component of e, so

$$e_i = c_i^T e = c_i^T M\epsilon.$$

Thus the requested representation is obtained by taking the coordinate vector c_i.

Final answer

$$\boxed{e_i = c_i^T(I_n - P)\epsilon, \qquad c_i = (0, \ldots, 0, 1, 0, \ldots, 0)^T.}$$

Part (b): Express the standardized residual as a ratio of quadratic forms

Step 1: Rewrite the standardized residual using RSS

Let $\nu = n - p - 1$. Since $\widehat{\sigma}^2_{UE} = \mathrm{RSS}/\nu$,

$$\frac{r_i^2}{\nu} = \frac{e_i^2}{(1 - p_{ii})\,\mathrm{RSS}}.$$

Step 2: Standardize the error vector

Under normality, write $\epsilon = \sigma Z$ with $Z \sim N_n(0, I_n)$. Then

$$e_i = \sigma c_i^T MZ, \qquad \mathrm{RSS} = \epsilon^T M\epsilon = \sigma^2 Z^T MZ.$$

Substituting,

$$\frac{r_i^2}{\nu} = \frac{(c_i^T MZ)^2}{(1 - p_{ii})Z^T MZ}.$$

Step 3: Write the numerator as a quadratic form

Since $M^T = M$,

$$(c_i^T MZ)^2 = Z^T Mc_i c_i^T MZ.$$

Define

$$Q = \frac{Mc_i c_i^T M}{1 - p_{ii}}.$$

Then

$$\boxed{\frac{r_i^2}{n - p - 1} = \frac{Z^T QZ}{Z^T MZ}.}$$

Part (c): Prove that Q is a projection matrix

Step 1: Compute the squared length of the vector Mc_i

Let $a = Mc_i$. Because M is symmetric and idempotent,

$$\begin{aligned} a^T a &= c_i^T M^T Mc_i \\ &= c_i^T M^2 c_i \\ &= c_i^T Mc_i \\ &= c_i^T (I - P)c_i \\ &= 1 - p_{ii}. \end{aligned}$$

Thus $1 - p_{ii} = a^T a > 0$ whenever the ith residual is not identically zero.

Step 2: Recognize Q as a rank-one orthogonal projector

With $a = Mc_i$,

$$Q = \frac{aa^T}{a^T a}.$$

This matrix is symmetric:

$$Q^T = \frac{(aa^T)^T}{a^T a} = \frac{aa^T}{a^T a} = Q.$$

It is also idempotent:

$$Q^2 = \frac{aa^T aa^T}{(a^T a)^2} = \frac{a(a^T a)a^T}{(a^T a)^2} = \frac{aa^T}{a^T a} = Q.$$

So Q is the orthogonal projection onto the one-dimensional space spanned by a.

Final answer

$$\boxed{Q^T = Q, \qquad Q^2 = Q, \qquad \operatorname{rank}(Q) = 1.}$$

Part (d): Independence and the Beta distribution

Step 1: Show that Q lies inside the residual projection

Because $a = Mc_i$ is in the column space of M, applying M to a leaves it unchanged: $Ma = a$. Hence

$$MQ = QM = Q.$$

Now define $R = M - Q$. Then

$$R^T = R,$$

and

$$R^2 = (M - Q)^2 = M^2 - MQ - QM + Q^2 = M - Q - Q + Q = M - Q = R.$$

Thus $R = I - P - Q$ is also an orthogonal projection.

Step 2: Find the ranks

We know $\operatorname{rank}(M) = n - p - 1 = \nu$, and $\operatorname{rank}(Q) = 1$. Since the range of Q lies inside the range of M,

$$\operatorname{rank}(R) = \operatorname{rank}(M) - \operatorname{rank}(Q) = \nu - 1 = n - p - 2.$$

Step 3: Prove that the two quadratic forms are independent

The projections are orthogonal because

$$QR = Q(M - Q) = Q - Q^2 = 0.$$

For a standard multivariate normal vector, quadratic forms corresponding to symmetric idempotent matrices with orthogonal ranges are independent. Therefore

$$U = Z^T QZ \sim \chi^2_1, \qquad V = Z^T RZ \sim \chi^2_{n-p-2}, \qquad U \perp\!\!\!\perp V.$$

Step 4: Express the denominator as the sum of the independent forms

Since $M = Q + R$,

$$Z^T MZ = Z^T QZ + Z^T RZ = U + V.$$

The ratio in part (b) is therefore

$$\frac{r_i^2}{n - p - 1} = \frac{U}{U + V}.$$

Step 5: Apply the chi-square ratio result

With $U \sim \chi^2_1$ and $V \sim \chi^2_{n-p-2}$ independent,

$$\frac{U}{U+V} \sim \text{Beta}\left(\frac{1}{2}, \frac{n-p-2}{2}\right).$$

Since $(n-p-2)/2 = \{(n-p)-2\}/2$, this is the Beta law resulting from one residual direction separated from the remaining $n-p-2$ directions. In the assignment's displayed parameterization, the second shape parameter is $(n-p-2)/2$.

Final answer

$$\boxed{\frac{r_i^2}{n-p-1} \sim \text{Beta}\left(\frac{1}{2}, \frac{n-p-2}{2}\right).}$$

The numerator is a χ^2_1 projection component; the remaining residual sum of squares is an independent χ^2_{n-p-2} component. Their ratio to their sum is Beta.

12 Question 12: Mean and variance of standardized residuals

Question (as assigned)

12. Let r_i, $i = 1, \ldots, n$, be the standardized residuals.

(a) Show that

$$r_i \overset{D}{=} \frac{\sqrt{n-p-1}\,T}{\sqrt{T^2+n-p-2}}, \qquad T \sim t_{n-p-2}.$$

(b) Show that the conditional distribution of T, given $T^2 = s$, is a two-point distribution with equal probability mass at $\pm\sqrt{s}$.

(c) Hence, or otherwise, show that $\mathbb{E}(r_i) = 0$.

(d) Show that $\text{var}(r_i) = 1$.

Required theory from the supplied lecture notes

From Question 11 and the supplied notes,

$$\frac{r_i^2}{n-p-1} \sim \text{Beta}\left(\frac{1}{2}, \frac{n-p-2}{2}\right).$$

The notes also give the externally studentized residual distribution

$$t_i = \frac{e_i}{\widehat{\sigma}_{UE,(-i)}\sqrt{1-p_{ii}}} \sim t_{n-p-2}.$$

A Student t_ν variable has a symmetric density about zero, and for $T \sim t_\nu$,

$$\frac{T^2}{T^2+\nu} \sim \text{Beta}\left(\frac{1}{2}, \frac{\nu}{2}\right).$$

The latter follows by writing $T = G/\sqrt{V/\nu}$ with $G^2 \sim \chi^2_1$, $V \sim \chi^2_\nu$ independently.

Let $\nu = n-p-1$, so the degrees of freedom of T in the question are $\nu - 1 = n-p-2$.

Part (a): Derive the t representation

Step 1: Use the independent normal and chi-square representation

The projection decomposition in Question 11 gives independent random variables

$$G \sim N(0,1), \qquad V \sim \chi^2_{\nu-1},$$

for which

$$r_i \overset{D}{=} \frac{\sqrt{\nu}\, G}{\sqrt{G^2+V}}.$$

Define

$$T = \frac{G}{\sqrt{V/(\nu-1)}}.$$

Then $T \sim t_{\nu-1}$. This definition preserves the sign of G and therefore gives the signed, not merely squared, representation.

Step 2: Algebraically rewrite the ratio using T

From the definition of T,

$$G = T\sqrt{\frac{V}{\nu-1}}.$$

Substitute this into the expression for r_i and cancel the common factor V under the square root:

$$\begin{aligned} r_i &\overset{D}{=} \frac{\sqrt{\nu}\, T\sqrt{V/(\nu-1)}}{\sqrt{T^2 V/(\nu-1)+V}} \\ &= \frac{\sqrt{\nu}\, T}{\sqrt{T^2+\nu-1}}. \end{aligned}$$

Since $\nu = n-p-1$, we have $\nu - 1 = n-p-2$. Therefore

$$\boxed{r_i \overset{D}{=} \frac{\sqrt{n-p-1}\, T}{\sqrt{T^2+n-p-2}}, \qquad T \sim t_{n-p-2}.}$$

Part (b): Conditional distribution given $T^2 = s$

Step 1: Use symmetry of the Student t distribution

The density of a Student t variable is even: $f_T(t) = f_T(-t)$. Once $T^2 = s$ is fixed with $s > 0$, the only possible values of T are $+\sqrt{s}$ and $-\sqrt{s}$.

Step 2: Compare the two conditional probabilities

By symmetry, the probability mass contributed by a small neighborhood around $+\sqrt{s}$ is equal to that around $-\sqrt{s}$. Therefore

$$\Pr(T = +\sqrt{s} \mid T^2 = s) = \Pr(T = -\sqrt{s} \mid T^2 = s) = \frac{1}{2},$$

understood as a regular conditional distribution (the event $T^2 = s$ has probability zero for a continuous variable).

Final answer

$$\boxed{\mathcal{L}(T \mid T^2 = s) = \tfrac{1}{2}\delta_{+\sqrt{s}} + \tfrac{1}{2}\delta_{-\sqrt{s}}.}$$

Part (c): Show that $\mathbb{E}(r_i) = 0$

Step 1: Observe that the transformation is odd

Define

$$g(t) = \frac{\sqrt{\nu}\, t}{\sqrt{t^2 + \nu - 1}}.$$

Then $g(-t) = -g(t)$. Since T has a symmetric distribution about zero, the positive and negative contributions cancel. Also $|g(T)| \leq \sqrt{\nu}$, so the expectation exists.

Final answer

$$\boxed{\mathbb{E}(r_i) = 0.}$$

Part (d): Show that $\mathrm{var}(r_i) = 1$

Step 1: Use the zero mean

Since $\mathbb{E}(r_i) = 0$,

$$\mathrm{Var}(r_i) = \mathbb{E}(r_i^2).$$

Step 2: Use the t representation

From part (a),

$$\mathbb{E}(r_i^2) = \nu\, \mathbb{E}\left(\frac{T^2}{T^2 + \nu - 1}\right).$$

For $T \sim t_{\nu-1}$,

$$\frac{T^2}{T^2 + \nu - 1} \sim \mathrm{Beta}\left(\frac{1}{2}, \frac{\nu - 1}{2}\right).$$

A $\mathrm{Beta}(a, b)$ variable has expectation $a/(a + b)$. Hence

$$\mathbb{E}\left(\frac{T^2}{T^2 + \nu - 1}\right) = \frac{1/2}{1/2 + (\nu - 1)/2} = \frac{1}{\nu}.$$

Step 3: Finish the variance calculation

Therefore

$$\mathrm{Var}(r_i) = \mathbb{E}(r_i^2) = \nu \cdot \frac{1}{\nu} = 1.$$

Final answer

$$\boxed{\mathbb{E}(r_i) = 0, \qquad \mathrm{Var}(r_i) = 1.}$$

The residual is bounded in magnitude by $\sqrt{n - p - 1}$; its mean is zero by symmetry and its second moment is one by the Beta expectation.

13 Question 13: Sensitivity of a contrast to a response perturbation

Question (as assigned)

13. Consider the normal linear regression model in (1). Suppose the Gram matrix $X^T X$ has an eigenvalue 0.001 with corresponding eigenvector

$$u = (1, -1, 0, \ldots, 0)^T.$$

Suppose that, due to a compilation error, the response of the last observation is perturbed by 0.1, while all predictor values remain unchanged. Thus,

$$Y^p = Y + (0, \ldots, 0, 0.1)^T, \qquad X^p = X.$$

Given that $x_{n,1} = 0.5$, determine the difference between the estimated contrast $\widehat{\beta}_0 - \widehat{\beta}_1$ obtained using the perturbed data and that obtained using the original data. Explain how the small eigenvalue of $X^T X$ influences the sensitivity of this contrast to the perturbation.

Required theory from the supplied lecture notes

The OLS estimator is

$$\widehat{\beta} = (X^T X)^{-1} X^T Y.$$

A perturbation ΔY with X unchanged changes the estimate by

$$\Delta\widehat{\beta} = (X^T X)^{-1} X^T \Delta Y.$$

If u is an eigenvector of $X^T X$ with eigenvalue $\lambda > 0$, then

$$(X^T X)^{-1} u = \lambda^{-1} u.$$

The supplied collinearity notes explain that small eigenvalues of $X^T X$ produce large inverse eigenvalues, so coefficient contrasts with a component in the corresponding eigendirection can be highly sensitive to small perturbations in the response.

Step 1: Represent the perturbation as a coordinate vector

Let e_n denote the nth coordinate vector in observation space. The response perturbation is

$$\Delta Y = 0.1 e_n.$$

The design matrix does not change, so

$$\begin{aligned} \widehat{\beta}^p - \widehat{\beta} &= (X^T X)^{-1} X^T (0.1 e_n) \\ &= 0.1 (X^T X)^{-1} x_n, \end{aligned}$$

where x_n is the nth design row written as a column vector.

Step 2: Write the requested contrast as a linear functional

The target is $\beta_0 - \beta_1$. Its contrast vector is

$$c = (1, -1, 0, \ldots, 0)^T = u.$$

Thus the change in the estimated contrast equals

$$\begin{aligned} \Delta(\widehat{\beta}_0 - \widehat{\beta}_1) &= c^T (\widehat{\beta}^p - \widehat{\beta}) \\ &= 0.1 c^T (X^T X)^{-1} x_n. \end{aligned}$$

Step 3: Use the small eigenvalue

Since $c = u$ and $(X^T X)u = 0.001u$, symmetry gives

$$u^T (X^T X)^{-1} = \frac{1}{0.001} u^T = 1000 u^T.$$

Therefore

$$\Delta(\widehat{\beta}_0 - \widehat{\beta}_1) = 0.1 \times 1000 \times u^T x_n.$$

Because the first column of X is the intercept, the first two coordinates of x_n are $(1, x_{n,1})^T$. Hence

$$u^T x_n = 1 - x_{n,1} = 1 - 0.5 = 0.5.$$

Step 4: Calculate the numerical difference

$$\Delta(\widehat{\beta}_0 - \widehat{\beta}_1) = 0.1 \times 1000 \times 0.5 = 50.$$

The perturbed-data estimate of the contrast is therefore 50 larger than the original-data estimate.

Final answer

$$\boxed{(\widehat{\beta}_0 - \widehat{\beta}_1)_{perturbed} - (\widehat{\beta}_0 - \widehat{\beta}_1)_{original} = 50.}$$

The eigenvalue 0.001 is inverted by the OLS formula, producing an amplification factor $1/0.001 = 1000$ in the direction $u = (1, -1, 0, \ldots, 0)^T$. The contrast $\beta_0 - \beta_1$ lies exactly in this weakly identified direction, so even a response change of only 0.1 produces a change of 50 in the estimated contrast.

14 Question 14: Determinant ratio after deleting multiple observations

Question (as assigned)

14. Let $(Y^{(-D)}, X^{(-D)})$ be a subset of (Y, X) after deleting a set D of d-many observations. Show that the ratio of determinants of the variances of $\widehat{\beta}$ and $\widehat{\beta}_{(-D)}$ can be expressed as $\det^{-1}(I - P(D))$, where $P(D)$ is the sub-matrix of $P(X)$ considering the indices corresponding to D.

Required theory from the supplied lecture notes

For a full-rank design under spherical errors,

$$\mathrm{Var}(\widehat{\beta}) = \sigma^2 (X^T X)^{-1}.$$

Deleting rows indexed by D subtracts their row outer products from the Gram matrix. The determinant identity

$$\det(I + AB) = \det(I + BA)$$

allows a determinant of a matrix whose dimension is the number of coefficients to be converted into a determinant of a matrix whose dimension is the number of deleted observations. This is the multi-observation form of the determinant identity used in the supplied covariance-ratio notes.

Notation. Let $m = p + 1$ be the number of columns of X. Let X_D be the $d \times m$ matrix of the deleted rows, and let $X_{(-D)}$ be the remaining design matrix. The principal submatrix of the full hat matrix corresponding to D is

$$P(D) = X_D (X^T X)^{-1} X_D^T.$$

Step 1: Write the Gram matrix after deletion

Every observation contributes one row outer product to $X^T X$. Removing the rows in D gives

$$X_{(-D)}^T X_{(-D)} = X^T X - X_D^T X_D.$$

Step 2: Form the determinant of the reduced Gram matrix

Factor out $X^T X$:

$$\begin{aligned} \det(X_{(-D)}^T X_{(-D)}) &= \det\{X^T X - X_D^T X_D\} \\ &= \det(X^T X)\det\{I_m - (X^T X)^{-1} X_D^T X_D\}. \end{aligned}$$

Step 3: Reduce the determinant to the deleted-observation space

Apply $\det(I - AB) = \det(I - BA)$ with

$$A = (X^T X)^{-1} X_D^T \in \mathbb{R}^{m \times d}, \qquad B = X_D \in \mathbb{R}^{d \times m}.$$

Then

$$\begin{aligned} \det\{I_m - (X^T X)^{-1} X_D^T X_D\} &= \det\{I_d - X_D (X^T X)^{-1} X_D^T\} \\ &= \det\{I_d - P(D)\}. \end{aligned}$$

Therefore

$$\boxed{\det(X_{(-D)}^T X_{(-D)}) = \det(X^T X)\det(I_d - P(D)).}$$

Step 4: Convert this into the ratio of covariance determinants

The full-sample covariance is

$$V_{full} = \mathrm{Var}(\widehat{\beta}) = \sigma^2 (X^T X)^{-1},$$

while the deleted-sample covariance is

$$V_{(-D)} = \mathrm{Var}(\widehat{\beta}_{(-D)}) = \sigma^2 (X_{(-D)}^T X_{(-D)})^{-1}.$$

Both are $m \times m$ matrices, so the scale determinant contributes $(\sigma^2)^m$ to each and cancels in their ratio:

$$\begin{aligned} \frac{\det(V_{(-D)})}{\det(V_{full})} &= \frac{\det(X_{(-D)}^T X_{(-D)})^{-1}}{\det(X^T X)^{-1}} \\ &= \frac{\det(X^T X)}{\det(X_{(-D)}^T X_{(-D)})} \\ &= \frac{1}{\det(I_d - P(D))}. \end{aligned}$$

Final answer

With the ratio oriented as deleted-sample covariance determinant divided by full-sample covariance determinant,

$$\boxed{\frac{\det\{\mathrm{Var}(\widehat{\beta}_{(-D)})\}}{\det\{\mathrm{Var}(\widehat{\beta})\}} = \det(I_d - P(D))^{-1}.}$$

This is the required result. The deleted observations affect coefficient precision through the determinant of the corresponding principal block of the hat matrix.

End of Problems Set 3 solutions